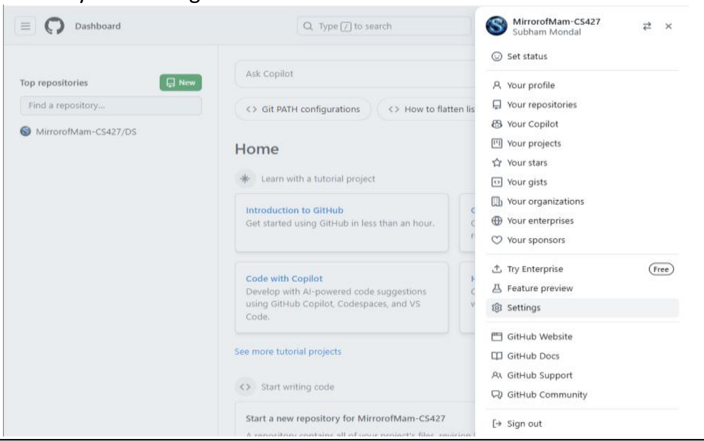
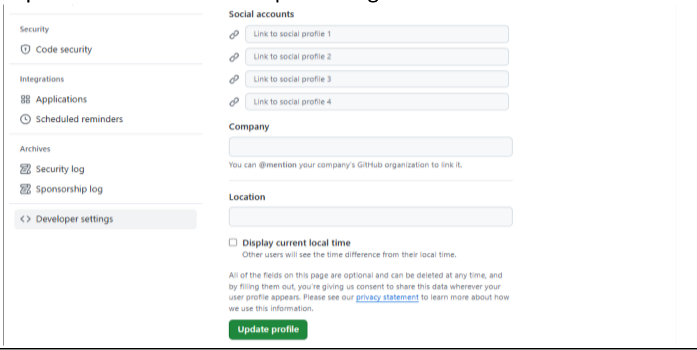
**Assignment 9:** Deploy a project from GitHub to EC2.

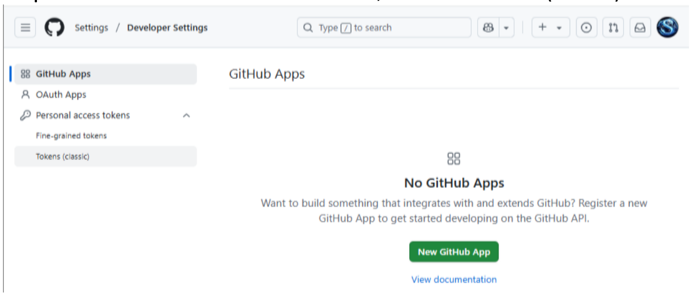
Step 1: After login into GitHub account, go to your profile section and click on your 'Settings'.



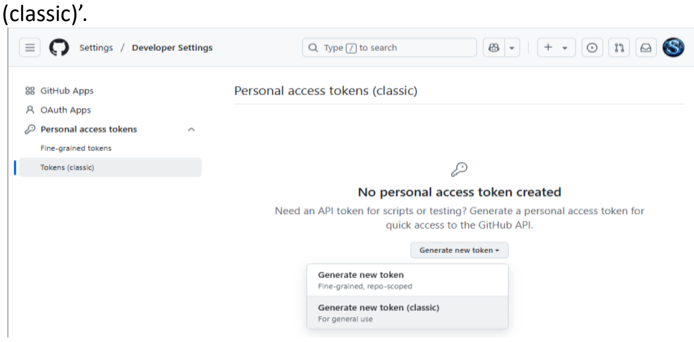

Step 2: Now click on 'Developer settings'.




Step 3: Under 'Personal access tokens', click on 'Tokens (classic)'.

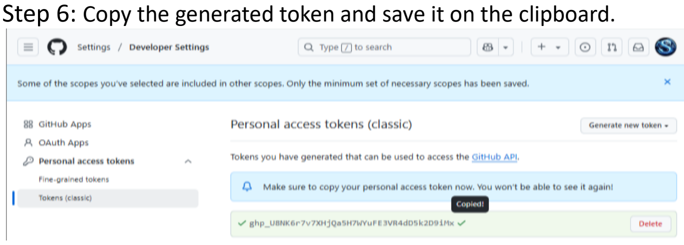



Step 4: Under 'Generate new token', click on 'Generate new token (classic)'.



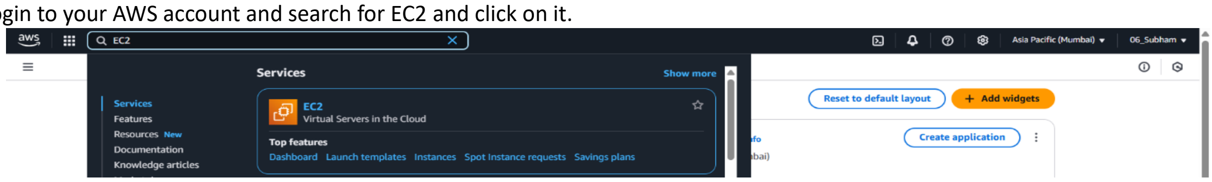

Step 5: Give a suitable Note, then tick on the check boxes then click on 'Generate token'.


Step 6: Copy the generated token and save it on the clipboard.

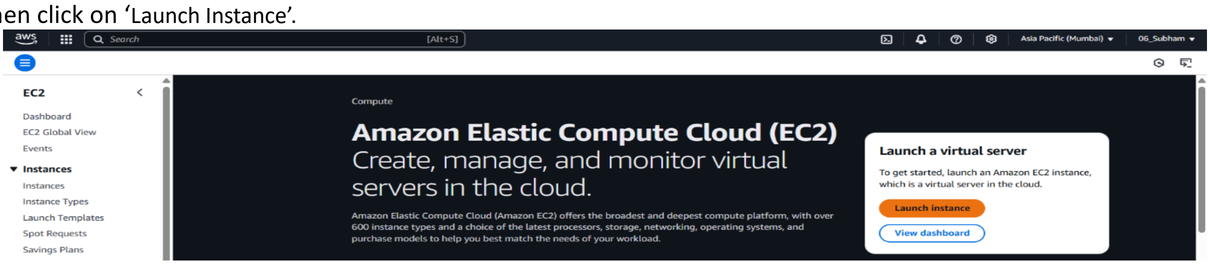



***Note: After that upload and commit a .js file along with .json file (index.js and Package.json).***

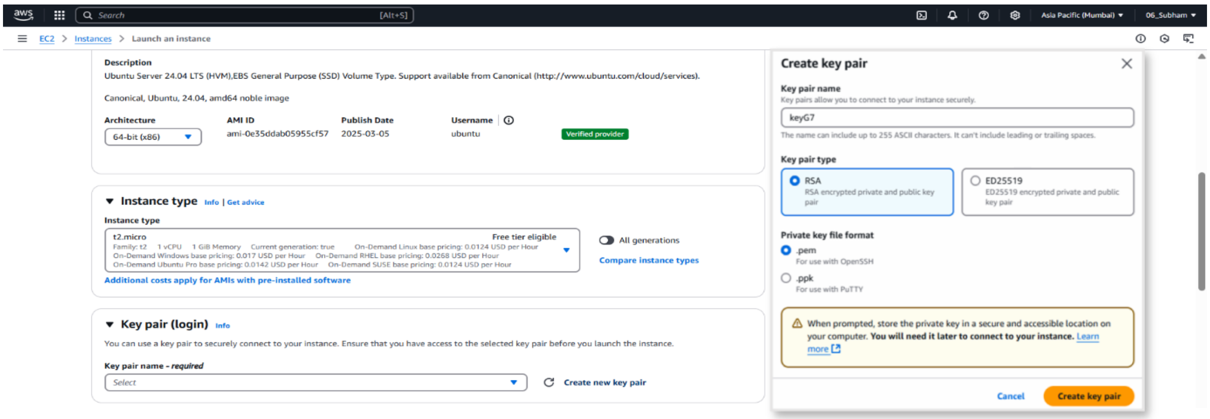
Step 7: Login to your AWS account and search for EC2 and click on it.



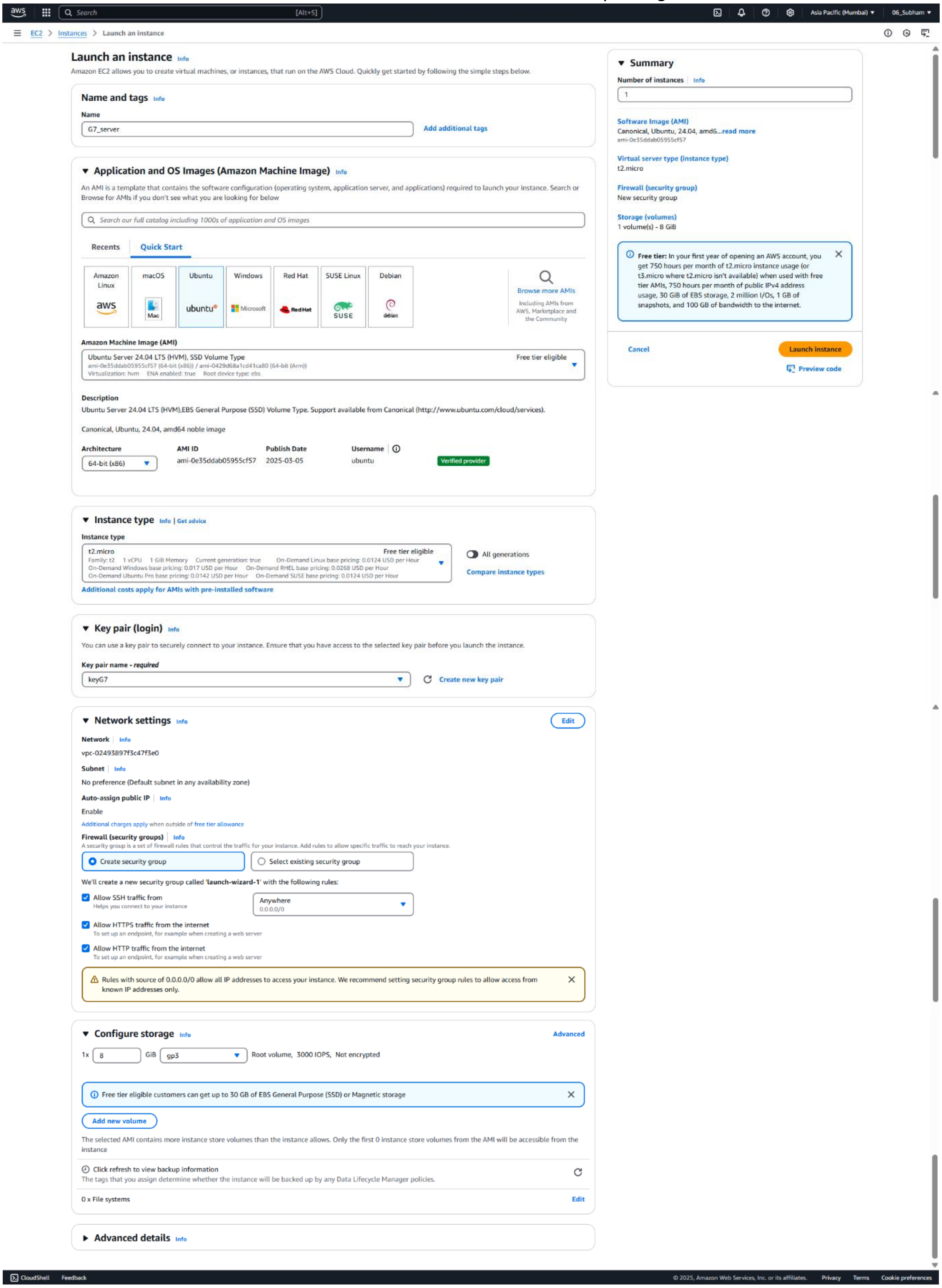

Step 8: Then click on 'Launch Instance'.



Step 9: Scroll down to ‘Key pair (login)’ and click on ‘Create new key pair’. A dialog box appears, fill the box as follows.



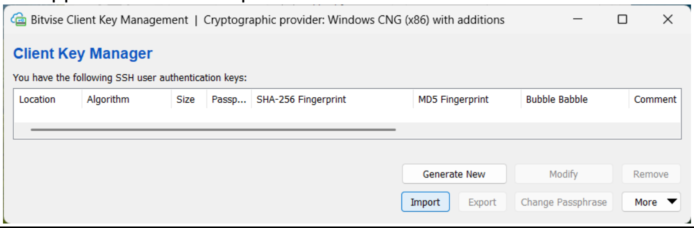

Step 10: Give a Name of the Instance and select Ubuntu as the OS and allow the necessary settings as follows. Then click on ‘Launch Instance’.



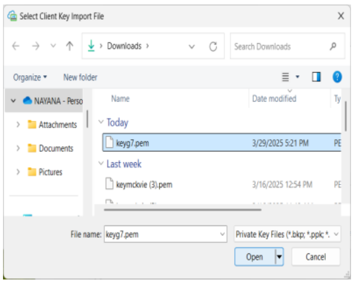

***Note: Click on the check box of the instance in the dashboard section, the details of the instance is visible.***

***Note: Now to create a connection between the instance (server) and the client, we need to use a third-party application Bitvise SSH Client. After installing the app a window will appears (like step 15 or 17).***

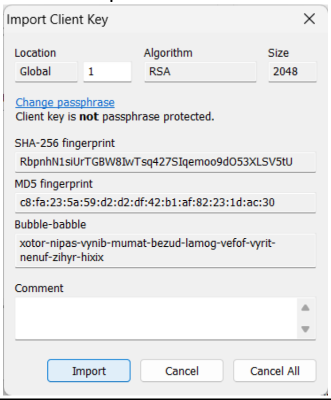
Step 11: From that click on 'Client key manager'. The following dialog box appears. Click on 'Import'.



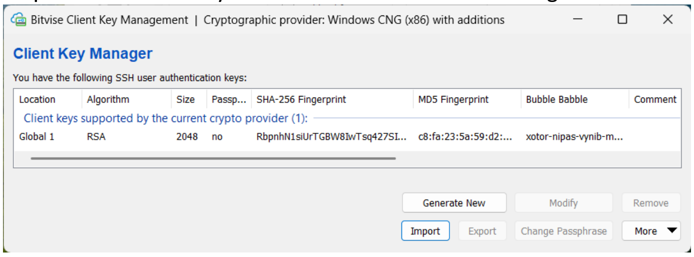

Step 12: Now select the file containing the client key and click on 'Open'.



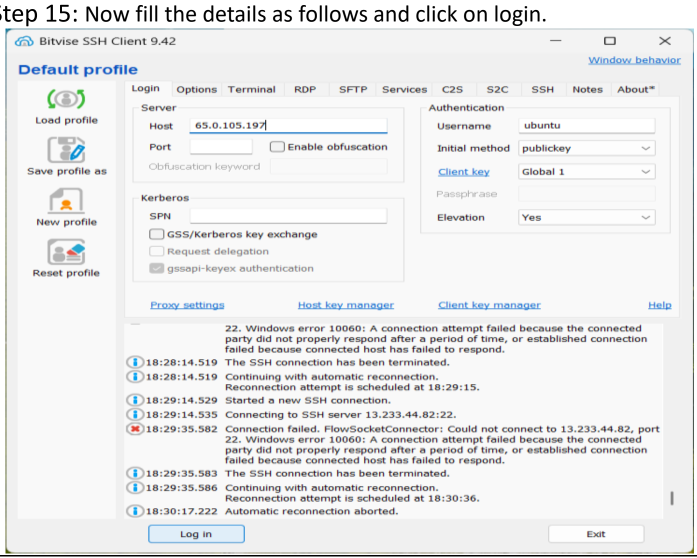
Step 13: The following appears, click on 'Import'.



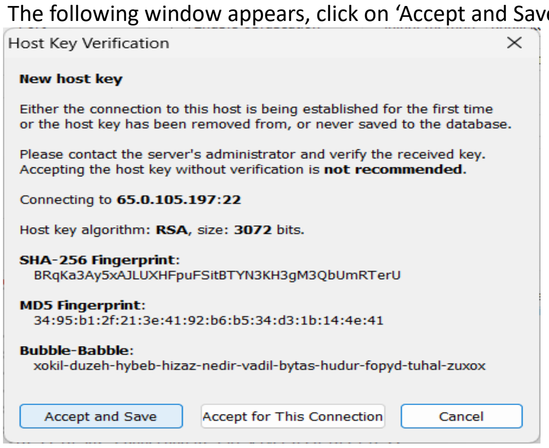

Step 14: The client key is now visible. Close the following window.



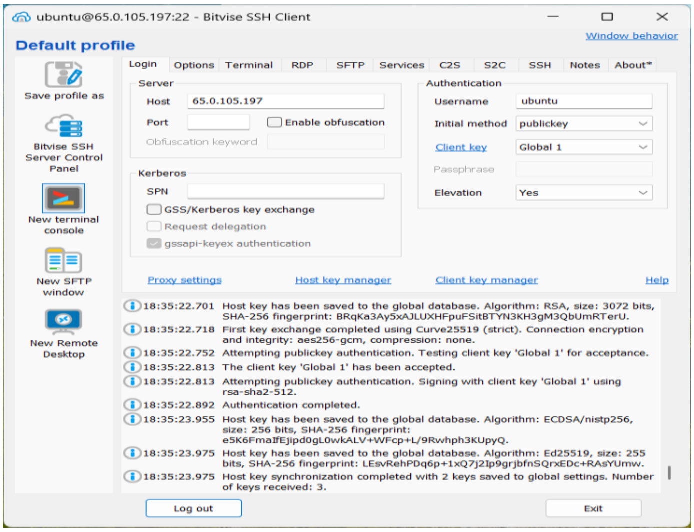

Step 15: Now fill the details as follows and click on login.



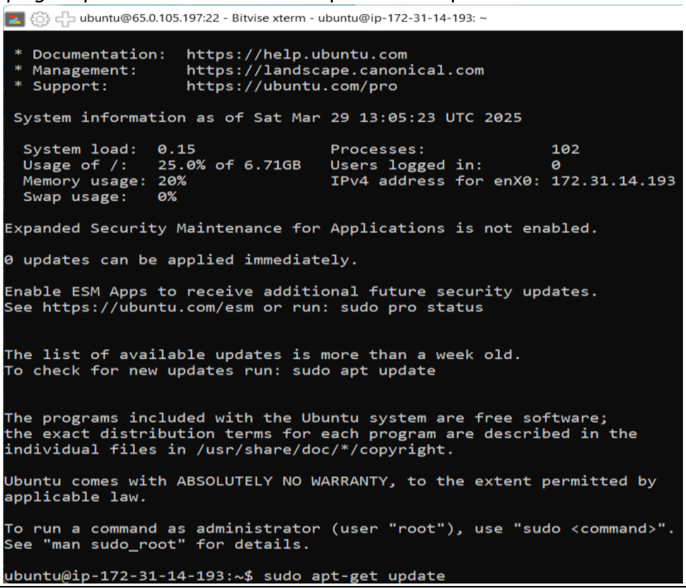

Step 16: The following window appears, click on 'Accept and Save'.



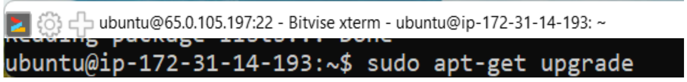

Step 17: The login process is successful. Now click on 'New terminal console' from the below window.

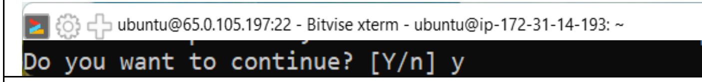



Step 18: Then a terminal will be open and run the command "*sudo apt-get update*". Then wait till the process completes.

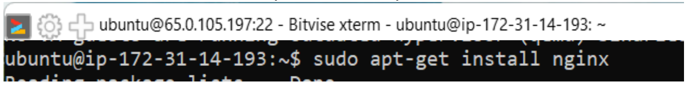


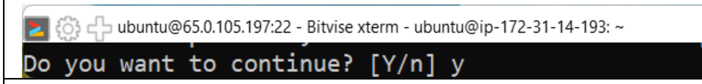

Step 19: Run the command "*sudo apt-get upgrade*". Then type yes the hit enter. Then wait till the process completes.



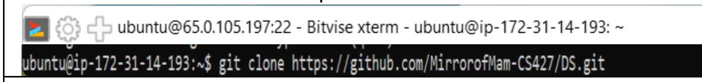




Step 20: Run the command "*sudo apt-get install nginx*". Type yes, the hit enters. Then wait till the process completes.

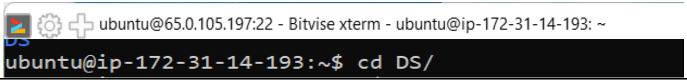


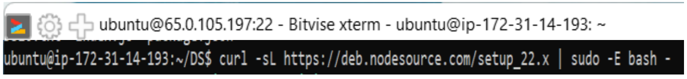

Step 21: Now run the command "*git clone <Repository link>*". In case of private GitHub repository, we need to login into the terminal using the GitHub user name and the personal access token.



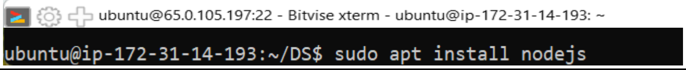

Step 22: Now run the command "*cd <Repository folder name>*".



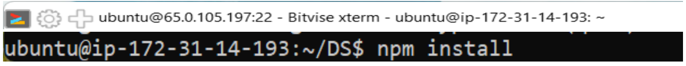

Step 23: Now run the command "*curl -sL https://deb.nodesource.com/setup_22.x | sudo -E bash –*".



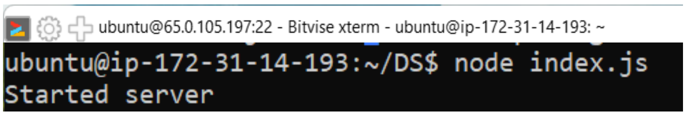
Step 24: Now run the command “*sudo apt install nodejs*”.



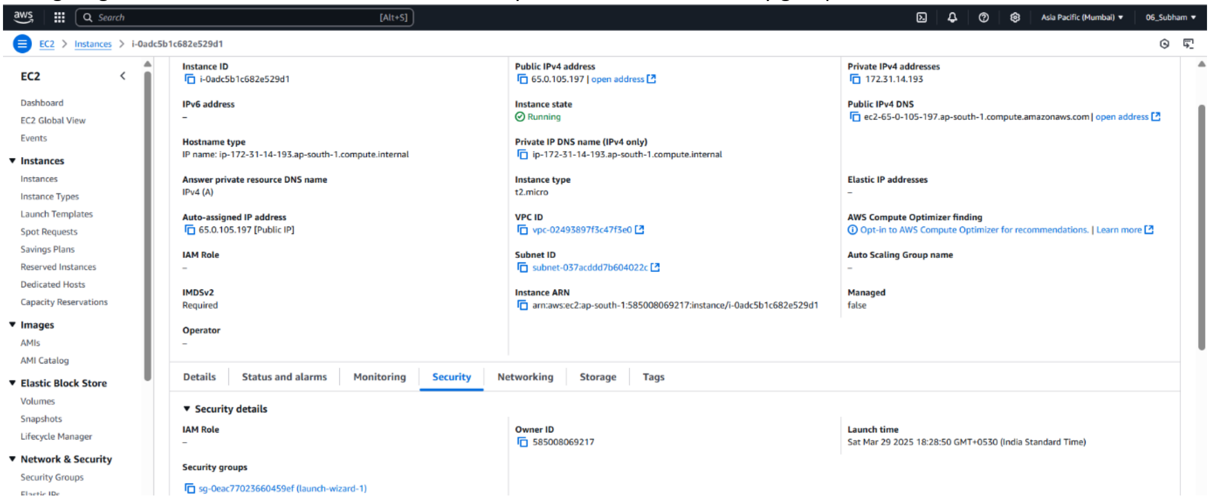

Step 25: Now run the command “*npm install*”.



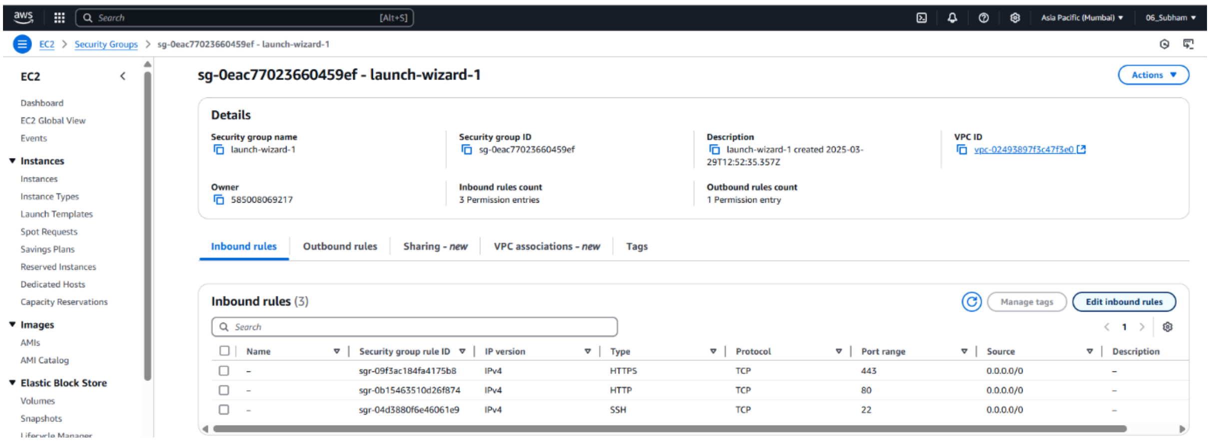

Step 26: Now give the command “*node <.js file>*” and press enter. The Server gets started.



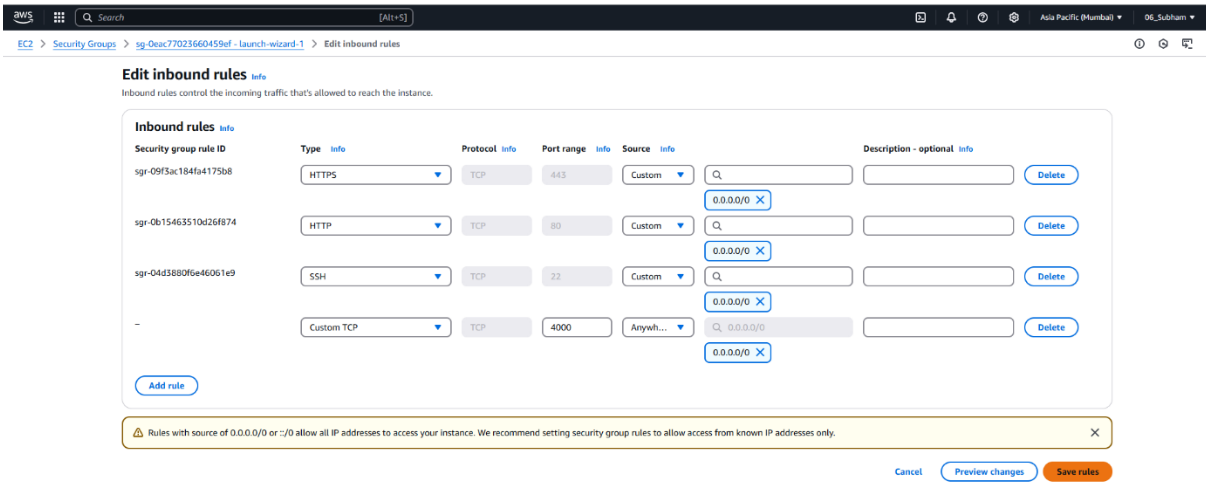

Step 27: Now again go to the EC2 instance and click on ‘Security’. Then under the Security group click on the blue coloured line.




Step 28: Click on ‘Edit inbound rules’.




Step 29: Click on ‘Add rule’. Give the rule type of “Custom TCP” & Port range at 4000 and rest of the fill as bellow image and click on ‘Save rules’.




Step 30: Open a browsing window and give the “*<instance IP address>:4000*” and hit enter. The web page (.js file) is seen.